

QUESTIONS

Q1
Harshad number
10 marks

Q2
GCD
10 marks

Q3
sum to a target value
10 marks

Q4
Bitwise Operators
10 marks

Q5
binary search
10 marks

5/5 answered

Q3 sum to a target value

10 marks

Complete the missing lines in the Python program below. The program finds all pairs of elements in a list that sum to a target value. Fill in all seven blanks.

```
def find_pairs(lst, target):  
    seen = _____ pairs = []  
    for num in _____:  
        complement = target - _____  
        if complement in _____:  
            pairs.append((complement, num))  
            _____add(num)  
    return _____  
data = [int(x) for x in input("Enter numbers: ").split()]  
t = int(input("Target: "))  
result = _____(data, t)  
print("Pairs:", result)
```

Your Answer

```
def find_pairs(lst, target):  
  
    seen = set()  
    pairs = []  
  
    for num in lst:
```

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5/5 answered

Q4 Bitwise Operators

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

Your Answer

```
a = 0b1101
b = 0o25
c = 0x1A
```

```
result = (a & b) | c
```

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Q1 Harshad number

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, $18 \% 9 == 0$. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
n=int(input())
s=0
t=n
while t>0:
    s+=t%10
    t//=10
```

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Q2 GCD

10 marks

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

```
a = int(input("Enter first: ")) b = int(input("Enter second: "))
while b = 0: temp = a a = b b = temp % a
print("GCD is", a)
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 a=int(input())
2 b=int(input())
3 while b!=0:
4     temp=b
5     b=a%b
6     a=temp
7 print("GCD:",a)
```

Test Input

```
12
6
```

Output

```
GCD: 6
```

⌚ Pending manual review

✓ Submitted